QQ7 Phenol V1

CIE 2022, OCR 2022, AQA 2022

IGC HK Exam

5 **(a)** Compare the relative acidities of benzoic acid (C_6H_5COOH), phenylmethanol ($C_6H_5CH_2OH$), and phenol (C_6H_5OH).
Explain your reasoning.

.... > .... > ....
most acidic   least acidic

....

....

....

....

....
[3]

(b) A series of nine separate experiments is carried out as shown in Table 5.1.

Complete the table by placing a tick (✓) in the relevant box if a reaction occurs. Place a cross (✗) in the box if no reaction occurs.

Table 5.1

	benzoic acid	phenylmethanol	phenol
Na(s)			
NaOH(aq)			
Na_2CO_3(aq)			

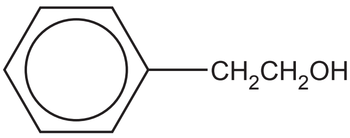
[3]

<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>5(a)</td><td>M1 benzoic acid > phenol > phenylmethanol

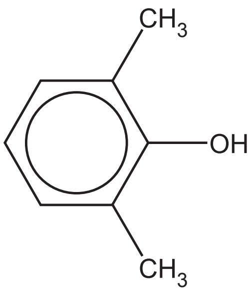
M2 / M3 Any two of:
• in benzoic acid negative inductive effect of C=O AND O-H bond is weakened
OR due to delocalisation of minus charge by C=O / 2O carboxylate ion is stabilised
• in phenol lone pair on oxygen is delocalised into the ring AND O-H bond is weakened
• in phenyl methanol positive inductive effect of CH_2 group AND O-H bond is strengthened</td><td>3</td></tr>
<tr><td>5(b)</td><td><table><tr><th></th><th>benzoic acid</th><th>phenylmethanol</th><th>phenol</th></tr><tr><td>Na(s)</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>NaOH(aq)</td><td>✓</td><td>✗</td><td>✓</td></tr><tr><td>Na_2CO_3(aq)</td><td>✓</td><td>✗</td><td>✗</td></tr></table>
Three correct for one mark, six correct for two marks, nine correct for three marks</td><td>3</td></tr>
</table>

20 This question is about the chemistry of aromatic compounds.

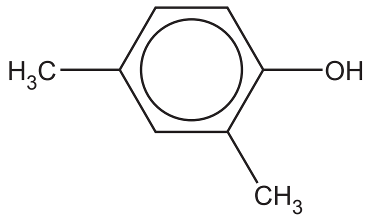
(a) Compounds **J**, **K** and **L**, shown below, are structural isomers.



Compound J



Compound K



Compound L

(i) What chemical test(s) could be used to confirm the presence of the phenol group in compounds **K** and **L**?

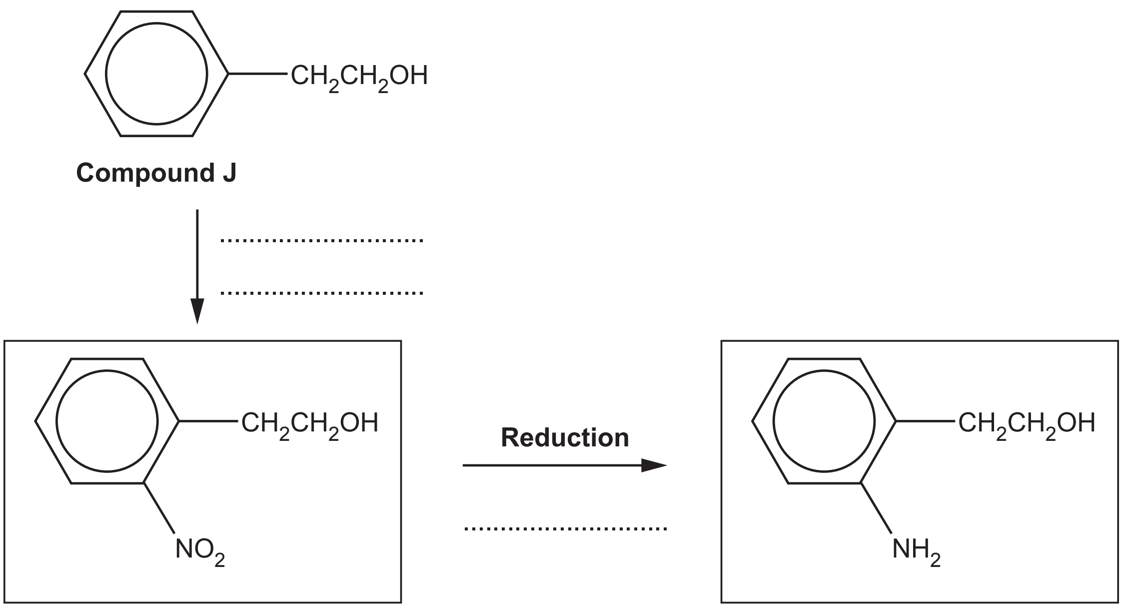
....

....

.... **[1]**

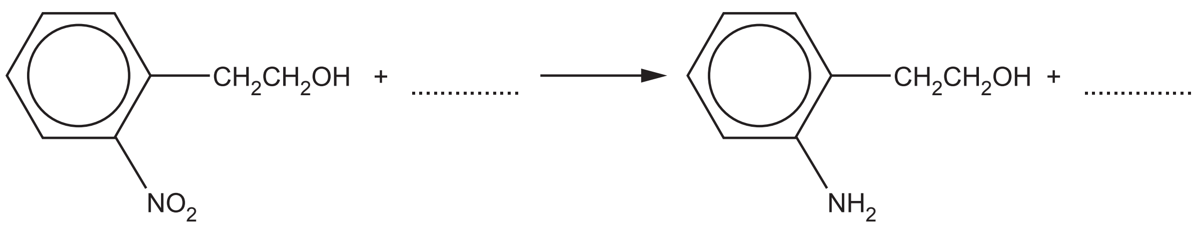
(d) A two-stage synthesis of an amine from compound **J** is shown below.

(i) Add the reagents for each stage of this synthesis.



[2]

(ii) Fill in the equation for the reduction stage of this synthesis.



[1]

(e) 1-phenylethanol is a naturally occurring compound found in many vegetables and flowers.

1-phenylethanol can be synthesised from 2-phenylethanol in two stages.

2-phenylethanol **1-phenylethanol**

Suggest reagents, conditions and equations for each stage in the synthesis.

Show structures for organic compounds.

Stage 1

reagents and conditions ....

equation:

Stage 2

reagents and conditions ....

equation:

[4]

(f) Acid anhydrides react in a similar way to acyl chlorides with phenols.

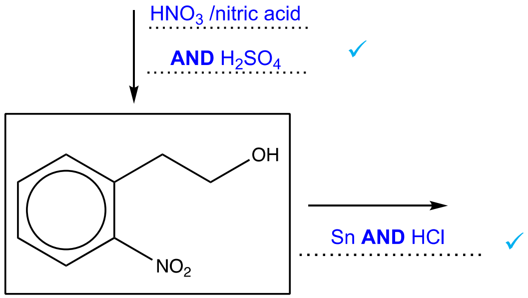
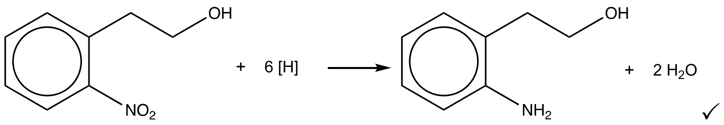
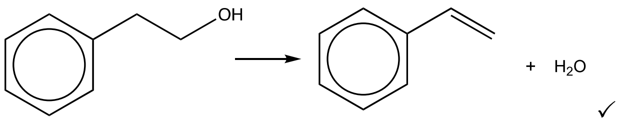
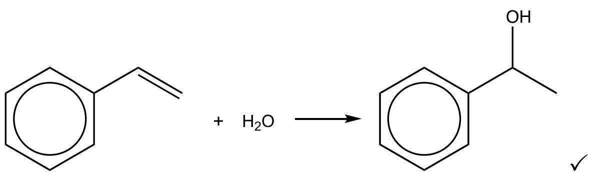
Benzoic anhydride is the acid anhydride of benzoic acid, C_6H_5COOH.

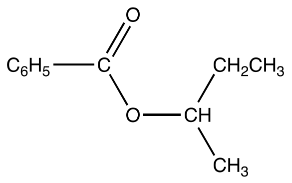
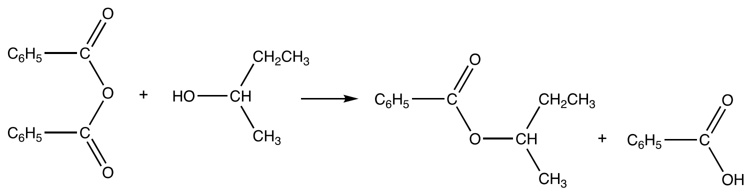
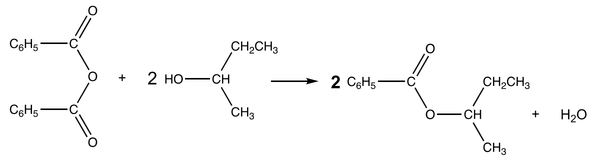
Benzoic anhydride reacts with butan-2-ol to form an ester.

Suggest an equation for this reaction. Show structures for organic compounds. Use C_6H_5 for any phenyl groups.

[2]

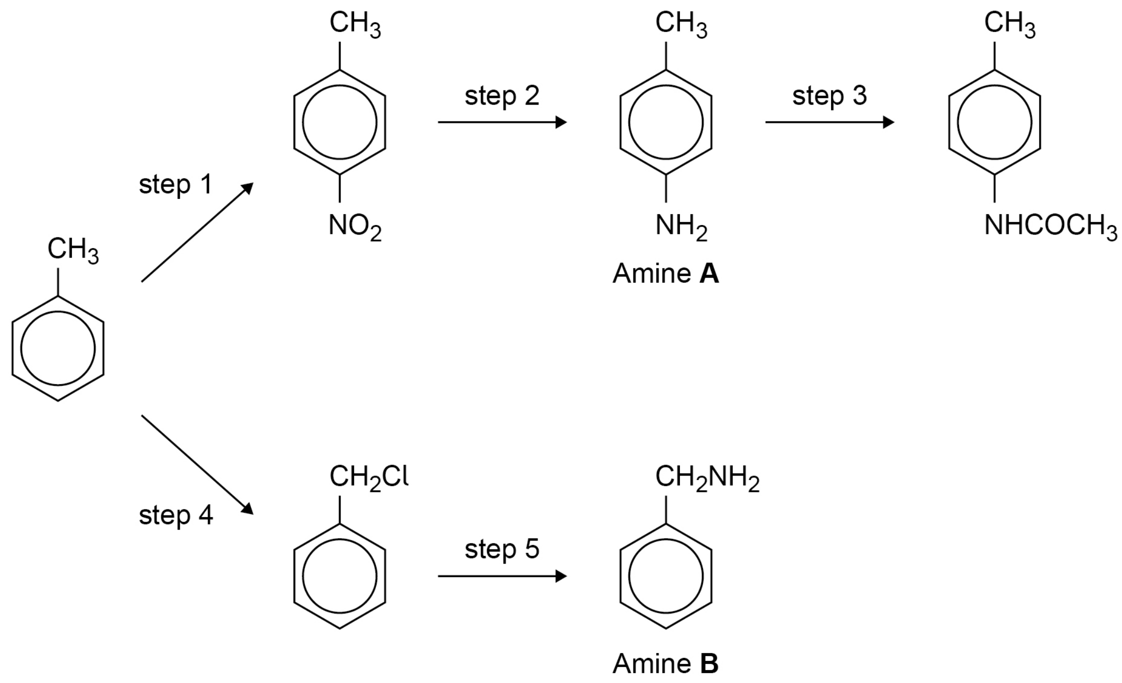
Question			Answer	Marks	AO element	Guidance
20	(a)	(i)	Indicator **AND** observation of acidity **AND** No reaction with carbonate ✓	1	AO1.2 ×1	**ALLOW** (Add) bromine **AND** white precipitate ✓ **ALLOW** (Add) $FeCl_3$ **AND** violet/purple colour ✓

20	(d)	(i)		2	AO1.2 ×2	**IGNORE** references to concentration **IGNORE** 'dilute' for HCl **IGNORE** H_2 **IGNORE** NaOH if seen as a reagent to convert nitro group into amine e.g 'Sn/(concentrated) HCl then NaOH' scores the mark
20	(d)	(ii)		1	AO2.6	
20	(e)		**Stage 1** *Reagents*: H_2SO_4 ✓  **Stage 2** *Reagents*: Steam/H_2O(g) **AND** acid/H^+ (catalyst) ✓ 	4	AO3.1 AO2.6 AO3.1 AO2.6	**ALLOW** any combination of skeletal **OR** structural **OR** displayed formula as long as unambiguous **ALLOW** H^+ **OR** HCl **OR** H_3PO_4 **DO NOT ALLOW** other named acids **IGNORE** concentration/pressure **IGNORE** water/steam For steam, **ALLOW** H_2O with temperature ≥100ºC **ALLOW** use of H_3PO_4/H_2SO_4 as catalyst **DO NOT ALLOW** HCl **IGNORE** pressure

20	(f)		Structure of ester product ✓  Correct balanced equation ✓ 	2	AO3.1 AO3.2	**ALLOW** any combination of skeletal **OR** structural **OR** displayed formula as long as unambiguous **ALLOW** 
			Total	22		

Do not write outside the box

1 0 This question is about the reaction scheme shown.



1 0 . 1 State the reagents needed for step **1** and the reagents needed for step **2**.

[3 marks]

step **1** ____

step **2** ____

1 0 . 2 Give the name of the mechanism for the reaction in step **3**.

[1 mark]

Do not write outside the box

1 0 . 3 Name the reagent for step **4**.

State a necessary condition for step **4**.

[2 marks]

Reagent ______________________________

Condition ______________________________

1 0 . 4 Amine **A** is formed in step **2** and amine **B** is formed in step **5**.

Explain why the yield of **B** in step **5** is less than the yield of **A** in step **2**.

[2 marks]

____

____

____

____

1 0 . 5 Explain why amine **B** is a stronger base than amine **A**.

[2 marks]

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10

END OF QUESTIONS

Question	Answers	Additional Comments/Guidelines	Mark
10.1	Step 1 Conc HNO_3 Step 1 Conc H_2SO_4 Step 2 Sn and HCl	 If conc missing in both allow 1 for HNO_3 and H_2SO_4 Allow Fe and HCl or Ni and H_2	M1 M2 M3 (3 x AO1)

Question	Answers	Additional Comments/Guidelines	Mark
10.2	(nucleophilic) addition-elimination		1 (AO1)

Question	Answers	Additional Comments/Guidelines	Mark
10.3	Chlorine UV (light)	Allow Cl_2 Allow sunlight / High temp (above 300^{o}c)	M1 M2 (2 x AO1)

Question	Answers	Additional Comments/Guidelines	Mark
10.4	In Step 5 further substitution / gives other amine products In Step 2 only one amine		1 1 (2 x AO3)

Question	Answers	Additional Comments/Guidelines	Mark
10.5	In B Alkyl group is electron donating or positive inductive effect Lone pair <u>on N</u> more available	Or in A lone pair (on N partially) delocalised Lone pair <u>on N</u> less available	1 1 (2 x AO2)